THE SUPPORT OF A FINITE TYPE QUASI-COHERENT MODULE IS CLOSED, DETECTED ON RESIDUE FIBRES, AND COMPATIBLE WITH PULLBACK

THE STACKS PROJECT AUTHORS

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Editorial note. Statement span 678-696 and proof span 698-726 come from the halfA pool card operations/pool-cards/stacks-morphisms-halfA.json and were re-read in the pinned chapter file before this card was written.

Editorial note. The proof refers to the three parts of the statement by number; those references are the author's and are preserved.

Editorial note. 7 cross-references occur inside the counted spans; they are listed in external_references as navigation pointers only.

Allowed standard prerequisites (editorial).

- The support of a finite type quasi-coherent module is closed, cited from the modules chapter, and the affine description of a finite type quasi-coherent module as the tilde of a finite module together with the closedness of supports over affine schemes. Cited at proof lines 915-920. Gives the closedness of the support, in the cited form or by combining the affine statements.
- Nakayama's lemma in the form that a stalk is zero exactly when its tensor product with the residue field is zero. Cited at proof lines 925-927. Supplies the second equivalence of part (2) and is used again at the end of the proof for the pullback.
- The stalk of a pullback module is the tensor product of the original stalk with the local ring, cited from the sheaves chapter, and the pullback of a finite type module is of finite type, cited from the modules chapter. Cited at proof lines 932-937. Reduces the compatibility of supports with pullback to the residue fibre criterion of part (2).

Source reference. Original lemma 00DV: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 512–512. Same-chapter reference used in the retained author text at source line(s) 709, 5128, 7341, 7915, 11427, 15408. It is supplied only as a navigation pointer so that the retained text compiles; the counted

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Source reference. Original lemma 00L2: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 772–772. Same-chapter reference used in the retained author text at source line(s) 705, 759, 11631, 11640, 11679, 11688. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 056I: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6711–6711. Same-chapter reference used in the retained author text at source line(s) 702. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 01B6: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1715–1715. Same-chapter reference used in the retained author text at source line(s) 712, 10678. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 01BA: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1719–1719. Same-chapter reference used in the retained author text at source line(s) 700, 5097, 7918. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 01PB: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2224–2224. Same-chapter reference used in the retained author text at source line(s) 703, 754, 4224, 5201, 5286, 6703, 10562, 11247. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 0098: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 344–344. Same-chapter reference used in the retained author text at source line(s) 720, 4907. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Lemma 0.1. *Let $\mathcal{F}$ be a finite type quasi-coherent module on a scheme X. Then*

(1) *The support of $\mathcal{F}$ is closed.*
(2) *For $x \in X$ we have*
$$x \in Supp(\mathcal{F}) \Leftrightarrow \mathcal{F}_x \neq 0 \Leftrightarrow \mathcal{F}_x \otimes_{\mathcal{O}_{X,x}} \kappa(x) \neq 0.$$
(3) *For any morphism of schemes $f : Y \to X$ the pullback $f^*\mathcal{F}$ is of finite type as well and we have $Supp(f^*\mathcal{F}) = f^{-1}(Supp(\mathcal{F}))$.*

Proof. Part (1) is a reformulation of Modules, Lemma 01BA. You can also combine Lemma 056I, Properties, Lemma 01PB, and Algebra, Lemma 00L2 to see this. The first equivalence in (2) is the definition of support, and the second equivalence

follows from Nakayama's lemma, see Algebra, Lemma 00DV. Let $f : Y \to X$ be a morphism of schemes. Note that $f^*\mathcal{F}$ is of finite type by Modules, Lemma 01B6. For the final assertion, let $y \in Y$ with image $x \in X$. Recall that

$$(f^*\mathcal{F})_y = \mathcal{F}_x \otimes_{\mathcal{O}_{X,x}} \mathcal{O}_{Y,y},$$

see Sheaves, Lemma 0098. Hence $(f^*\mathcal{F})_y \otimes \kappa(y)$ is nonzero if and only if $\mathcal{F}_x \otimes \kappa(x)$ is nonzero. By (2) this implies $x \in \text{Supp}(\mathcal{F})$ if and only if $y \in \text{Supp}(f^*\mathcal{F})$, which is the content of assertion (3). □